

Script Breakdown Tool

Build Spec Sheet — Version 2

Updated Output Schema · Industry-Standard Budget Structure · Scheduling Placeholders

1. OVERVIEW & WHAT HAS CHANGED

What This Spec Sheet Does

This is Version 2 of the build specification for the Script Breakdown Tool. It supersedes the original spec (Tool 1 of 2) in one important way: the output spreadsheet structure has been completely redesigned to align with the industry-standard Movie Magic Budgeting account numbering system. Everything else — the CLI interface, the Claude API call, the PDF parsing, the openpyxl generation, the formula-only math rule — remains exactly the same as Version 1.

The Core Insight Driving This Change

The original output organized line items by production department tabs (Props, Wardrobe, Camera, etc.). This is a reasonable breakdown structure but it is NOT the structure a Line Producer uses when building a budget. The industry standard — used by Movie Magic Budgeting and every professional film budget — organizes costs by numbered accounts. The new output puts line items directly into those accounts, with script-extracted data populating the fields that a script breakdown can answer, and leaving all other fields blank as clearly-labeled placeholders for future tools or manual entry.

KEY PRINCIPLE: The spreadsheet output is the handoff contract between the breakdown tool (Tool 1) and the future pricing tool (Tool 2). Getting this structure right now means Tool 2 can be built to slot directly into the same sheet without restructuring. Do not deviate from the account numbering system defined in this spec.

What the Script Breakdown Can and Cannot Answer

Not every budget account can be populated from a script. Claude must understand which fields it is responsible for filling and which must remain blank. The three categories are:

- **SCRIPT-POPULATED** — Claude extracts this directly from the script PDF. These cells are filled by the tool.
- **SCHEDULE-PLACEHOLDER** — Requires a shooting schedule (stripboard/Day-Out-of-Days). Leave blank with a light gray fill and the label [SCHEDULE]. A future scheduling tool or the 1st AD fills these manually.
- **DEAL-PLACEHOLDER** — Requires negotiated crew/cast contracts. Leave blank with the label [DEAL]. The Line Producer fills these.
- **FORMULA** — Never hardcode. Always write as an Excel formula referencing other cells.

2. ARCHITECTURE — UNCHANGED FROM V1

The pipeline is identical to Version 1. Only the output structure changes.

Step	Action
1	User runs: <code>python breakdown.py --input script.pdf --output budget.xlsx</code>
2	Python reads the PDF and sends it to Claude API as base64-encoded document
3	Claude returns strictly-formatted JSON — no prose, no markdown fences, just data
4	Python parses the JSON and writes it into a formatted .xlsx using openpyxl
5	All math cells (Subtotal, Total) are written as Excel formulas — never hardcoded
6	User opens the .xlsx — no AI interface, no chat window. A spreadsheet.

Technical Requirements — Unchanged

Component	Requirement
Language	Python 3.10+
AI SDK	anthropic — PDF sent as base64 document in API call
Spreadsheet	openpyxl — all Excel generation
Interface	argparse CLI — no GUI, no chat interface
AI output	Strict JSON only — Claude prompt must forbid prose
Modularity	Spreadsheet writer must be a separate function (swap to Google Sheets later)
Math rule	NEVER hardcode a calculated value. All totals = Excel formula strings

3. NEW SPREADSHEET STRUCTURE

Tab Structure Overview

The new workbook has three types of tabs. Every tab except the Top Sheet and Scene Breakdown uses the identical column structure defined below.

Tab Name	Purpose	Who Populates It
TOP SHEET	One-line-per-account summary. Grand total. First thing a producer sees.	Formula — auto-sums all account tabs
SCENE BREAKDOWN	One row per scene. Script elements tagged by color-coded category. Source traceability.	Claude — fully populated from script
1100 – Development	Story rights, script, legal, office overhead pre-production.	DEAL-PLACEHOLDER (no script data)
1200 – Story & Rights	Story purchase, writer fees, continuity.	DEAL-PLACEHOLDER
1400 – Producers Unit	EP, Producer, Line Producer fees.	DEAL-PLACEHOLDER
1500 – Directors Unit	Director fee, assistant, storyboard.	DEAL-PLACEHOLDER
1600 – Talent (Cast)	Principal cast, supporting, day players, stunt performers, extras casting.	SCRIPT populates character names & scene count. Rates = DEAL-PLACEHOLDER.
2100 – Production Staff	UPM, 1st AD, 2nd AD, coordinators, auditors, location manager.	SCHEDULE + DEAL (crew days x rate)
2200 – Art Direction	Production designer, art director, sketch artists, materials.	SCRIPT flags complexity. Crew = DEAL.
2300 – Set Construction	Construction coordinator, labor, materials, stage rental.	SCRIPT populates set builds needed. Labor = DEAL.
2400 – Set Decoration	Set decorator, swing gang, dressings purchased/rented.	SCRIPT populates set dressing needs. Crew = DEAL.
2500 – Props	Property master, assistant, picture vehicles, animals, armorer, firearms.	SCRIPT-POPULATED (richest account from breakdown)
2600 – Camera	DP, operators, assistants, camera package, special rentals (drone, underwater, aerial).	SCRIPT flags special rigs. Package = DEAL.
2700 – Electric	Gaffer, best boy, electricians, generator, lighting package.	SCRIPT flags night/special lighting. Crew = DEAL.
2800 – Grip	Key grip, dolly grip, crane, grip package.	SCRIPT flags crane/dolly needs. Package = DEAL.
2900 – Production Sound	Mixer, boom operator, sound package, walkie-talkies.	SCRIPT flags dialogue complexity. Package = DEAL.
3000 – Mechanical FX	Practical effects: rain, fire, fog, explosions, pyrotechnics, snow.	SCRIPT-POPULATED
3100 – Special VFX	Visual effects supervisor, VFX shots, miniatures, mattes.	SCRIPT-POPULATED (shot count estimated)
3200 – Set Operations	Set carpenter, standby painter, craft service, first aid, security.	SCHEDULE-driven. Script flags first aid needs.
3300 – Wardrobe	Costume designer, costumers, manufacture, rentals, purchases.	SCRIPT-POPULATED (period, character count, special)
3400 – Makeup & Hair	Head makeup, hair stylists, special makeup design, prosthetics.	SCRIPT-POPULATED (notable only — prosthetics, aging, wounds)
3500 – Locations	Survey, travel, site rentals, permits, catering, guards, police.	SCRIPT-POPULATED (location count, type, permit flags)

3600 – Transportation	Transport coordinator, drivers, picture vehicles, production vehicles.	SCRIPT populates picture vehicles. Crew = SCHEDULE.
3800 – Studio Facilities	Stage rental, back lot, electric power, heating/AC.	SCRIPT flags stage vs. practical. Rates = DEAL.
3900 – Atmosphere	Background extras, stand-ins, silent bits, minors.	SCRIPT-POPULATED (extras count per scene)
5100 – Editing	Post prod supervisor, editor, assistant editors, editing rooms.	DEAL-PLACEHOLDER (post, not from script)
5200 – Post Film/Lab	Color, negative cutting, answer prints. (Digital equiv: DI, color grade, deliverables.)	DEAL-PLACEHOLDER
5300 – Post Sound	ADR, Foley, sound design, dubbing, mixing.	DEAL-PLACEHOLDER
5400 – Music	Composer, musicians, rights, licensing.	DEAL-PLACEHOLDER
5500-5700 – Titles/Opticals/Post Video	Titles, opticals, VFX post, digital intermediate.	DEAL-PLACEHOLDER
6100 – Insurance	Cast insurance, negative, E&O;, liability.	DEAL-PLACEHOLDER (% of budget)
6200 – Legal	Legal fees, contracts, clearances.	DEAL-PLACEHOLDER
6300 – Publicity	Publicist, EPK, stills, press.	DEAL-PLACEHOLDER
6400 – Miscellaneous	Accounting, bank charges, contingency (10%), completion bond.	FORMULA (contingency = 10% of total)

4. UNIVERSAL COLUMN STRUCTURE — ALL ACCOUNT TABS

Every account tab (1100 through 6400) uses the identical column layout below. This is the integration contract — Tool 2 (pricing) will write into columns F and H. Do not change column letters without updating both tools.

Co l	Field Name	Data Type	Populated By	Notes
A	Account No	Text	Python (hardcoded per tab)	e.g. '2500.01', '2500.02' — sub-account within the tab
B	Description	Text	Claude or template label	Line item name. Template rows pre-filled with standard MM labels. Claude adds extracted rows.
C	Amount	Numeric	Claude or [SCHEDULE]/[DEAL]	How many: people, days, units. SCRIPT rows filled. Others left blank with placeholder label.
D	Unit Type	Text	Claude or [SCHEDULE]/[DEAL]	Weeks / Days / Hours / Flat / Allow / Rental-Day / Rental-Week / Purchase
E	Rate	Numeric — BLANK	[DEAL] or Tool 2	Cost per unit. LEFT BLANK. Tool 2 or LP fills this. Never estimated by Claude.
F	Subtotal	Formula	Excel	=C*D*E — written as formula string by openpyxl. Never hardcoded.
G	Fringe %	Numeric — BLANK	[DEAL]	Tax/benefit load on labor. LEFT BLANK. LP enters applicable union fringe rate.
H	Total	Formula	Excel	=F+(F*G) — written as formula string. Never hardcoded.
I	Vendor	Text — BLANK	Tool 2	Populated by pricing tool. LEFT BLANK now.
J	Vendor URL	Hyperlink — BLANK	Tool 2	Populated by pricing tool. LEFT BLANK now.
K	Script Page	Hyperlink	Claude	Page number in source PDF. Written as clickable hyperlink to source PDF using openpyxl write_url().
L	Script Quote	Text	Claude	Verbatim excerpt (max 20 words) from script that triggered this line item.
M	Notes	Text	Claude	Additional context, flags, or production notes extracted from script.
N	Confidence	Text	Claude	High / Medium / Low — AI certainty about this extraction.
O	Populated By	Text	Python	SCRIPT / SCHEDULE / DEAL / FORMULA — color-coded. Tells producer where data came from or needs to come from.

*FORMULA RULE: Column F must always be written as the string '=C{row}*D{row}*E{row}'. Column H must always be written as '=F{row}+(F{row}*G{row})'. Use openpyxl to write these as formula strings — never compute the value in Python and hardcode it. If a cell has no formula (e.g. a blank placeholder row), leave it empty.*

Placeholder Row Formatting

For every standard MM budget line item that Claude cannot populate from the script, Python must still write the row with the standard description label in column B, the appropriate placeholder in column C ([SCHEDULE] or [DEAL]), and leave columns E, G, I, J blank. Apply a light gray fill (hex #f1f3f5) to placeholder rows so the producer can distinguish auto-populated rows from manual-entry rows at a glance.

Color Coding for Column O (Populated By)

Value	Cell Fill Color	Meaning
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SCRIPT	#d4edda (light green)	Claude extracted this from the script. Data is present.
SCHEDULE	#fff3cd (light yellow)	Requires shooting schedule. 1st AD must fill this.
DEAL	#f8d7da (light red/pink)	Requires negotiated deal. Line Producer must fill this.
FORMULA	#d1ecf1 (light blue)	Calculated automatically. Do not edit.

5. SCENE BREAKDOWN TAB — DETAILED SPEC

The Scene Breakdown tab replaces the old per-department tabs as the primary script-extraction output. It is one row per scene, not one row per line item. It gives producers a bird's-eye view of every scene and its cost drivers before diving into account detail tabs.

Col	Field	Notes
A	Scene Number	From script heading. e.g. '14', '14A'
B	Scene Heading	Full slug line: INT./EXT. LOCATION – DAY/NIGHT
C	Int / Ext	INT or EXT — extracted from slug
D	Day / Night	DAY or NIGHT — extracted from slug. Night = cost flag.
E	Page Count (eighths)	Script pages in eighths. e.g. '4/8' = half page. Estimate if not explicit.
F	Script Day	Story day number (not shooting day). e.g. 'Day 1', 'Day 3 — Evening'
G	Cast (Red)	Comma-separated character names appearing in scene
H	Extras / Background (Yellow)	Estimated extras count. Note: 'crowd', 'diner regulars (8)', etc.
I	Props (Green)	Key props flagged in scene — hero props, breakaways, weapons
J	Practical FX (Blue)	On-set practical effects: fire, rain, fog, explosions, pyro
K	Wardrobe (Orange)	Notable wardrobe: period, special, quick-change, stunts
L	Animals (Purple)	Any livestock, trained animals, creatures
M	Vehicles / Picture Cars (Brown)	On-screen vehicles — type and count
N	Special Makeup (Pink)	Prosthetics, aging, wounds, non-standard makeup only
O	Special Camera (Teal)	Crane, drone, Steadicam, underwater, aerial, high-speed
P	Stunts (Gray)	Stunt action, fights, falls, car work, wire work
Q	VFX (Cyan)	Estimated VFX shot count + brief description
R	Location Type	Practical (real place) / Stage (build) / Unclear
S	Shooting Day	[SCHEDULE] — left blank. 1st AD fills from stripboard.
T	Estimated Shoot Hours	[SCHEDULE] — left blank. 1st AD fills.
U	Notes / Flags	Any production flags: minors on set, weapons, permits required, fire safety, etc.
V	Script Page Ref	Hyperlink to PDF page where scene begins

COLOR CODING NOTE: The industry color-coding system (Red=Cast, Yellow=Extras, Green=Props, Blue=SFX, Orange=Wardrobe, Purple=Animals, Brown=Vehicles, Pink=Makeup, Teal=Camera, Gray=Stunts) should be applied as Excel cell background fills in the Scene Breakdown tab column headers — matching the standard colors a 1st AD would recognize from Movie Magic Scheduling breakdown sheets.

6. TOP SHEET TAB — SPEC

The Top Sheet is the first tab in the workbook and the first thing a producer sees. It is a one-page summary — one row per major account group, auto-calculated from the detail tabs using SUM formulas. No data is entered here manually.

Section	Account Range	Label	Total Formula
ABOVE THE LINE	1100–1200	Development + Story & Rights	=SUM('1100'!H:H, '1200'!H:H)
	1400	Producers Unit	=SUM('1400'!H:H)
	1500	Directors Unit	=SUM('1500'!H:H)
	1600	Talent / Cast	=SUM('1600'!H:H)
		TOTAL ABOVE THE LINE	=SUM(above ATL rows)
BELOW THE LINE	2100	Production Staff	=SUM('2100'!H:H)
	2200–2400	Art Direction + Set Construction + Set Decoration	=SUM(...)
	2500	Props	=SUM('2500'!H:H)
	2600	Camera	=SUM('2600'!H:H)
	2700	Electric	=SUM('2700'!H:H)
	2800	Grip	=SUM('2800'!H:H)
	2900	Production Sound	=SUM('2900'!H:H)
	3000–3100	Mechanical FX + Special VFX	=SUM(...)
	3200	Set Operations	=SUM('3200'!H:H)
	3300	Wardrobe	=SUM('3300'!H:H)
	3400	Makeup & Hair	=SUM('3400'!H:H)
	3500	Locations	=SUM('3500'!H:H)
	3600	Transportation	=SUM('3600'!H:H)
	3800–3900	Studio Facilities + Atmosphere (Extras)	=SUM(...)
		TOTAL BELOW THE LINE	=SUM(all BTL rows)
POST PRODUCTION	5100–5700	Editing, Post Film/Lab, Sound, Music, Titles, VFX Post	=SUM(...)
		TOTAL POST PRODUCTION	=SUM(all post rows)
OTHER	6100	Insurance	=SUM('6100'!H:H)
	6200	Legal	=SUM('6200'!H:H)
	6300	Publicity	=SUM('6300'!H:H)
	6400	Contingency (10%)	=(TOTAL_ATL+TOTAL_BTL+TOTAL_POST)*0.10
		GRAND TOTAL	=SUM(ATL+BTL+POST+OTHER)

7. WHICH ACCOUNTS CLAUDE POPULATES — DETAILED RULES

The following accounts are where Claude adds actual extracted rows based on script content. All other accounts exist in the workbook as template shells (standard MM line item labels pre-filled) but with [DEAL] or [SCHEDULE] placeholders in column C.

1600 — Talent / Cast

- Extract every named speaking character. One row per character.
- Column B: Character name. Column C: number of scenes they appear in (from Scene Breakdown tab count). Column D: 'Days'. Column E: [DEAL].
- Add rows for: Featured Extras (if described specifically), Stand-ins (one per lead), Stunt Doubles (if stunt action involves a named character).
- Add one row for Background/Extras casting fee as Allow if any crowd scenes exist.
- Add one row for Welfare Worker/Teacher if any minor characters appear.

2200 / 2300 / 2400 — Art Direction / Construction / Set Decoration

- For each distinct location in the script: add a row in 2400 (Set Decoration) for 'Set Dressing — [Location Name]'. Column C = 1, Unit = Allow.
- If a script location implies construction (spaceship, fantasy environment, period set not findable as a practical): add a row in 2300 (Set Construction) flagged High priority.
- Period settings (any specific decade or era in the scene heading): add rows for 'Period Set Dressing' in 2400 and 'Period Props' in 2500.
- Flag with Confidence=Medium if it's unclear whether the location is practical or a build.

2500 — Props (RICHEST ACCOUNT)

- This is the account most richly populated by the script. Extract every prop explicitly named or clearly implied in action lines.
- Hero props (handled by cast on camera): Unit Type = Purchase, Quantity = 2 minimum (one hero, one backup).
- Breakaway props (must be destroyed on camera): Unit Type = Purchase, Quantity = 3 minimum.
- Weapons (any firearm, knife, sword, edged weapon): add both a Props row AND a Stunts & Safety row for armorer.
- Animals: add row in 2500 for animal rental + handler. Licensed wrangler legally required — note in Notes column.
- Picture Vehicles: add row in 2500 for each on-screen vehicle type. Also add corresponding row in 3600.
- Firearms: add separate rows for 'Prop firearms' and 'Blank ammunition' in account 2500.

2600 — Camera Operations

- Standard camera package: always add one row for 'Main Camera Package' — Amount=1, Unit=Rental-Day, Rate=[DEAL].
- Handheld rig: add if script specifies 'HANDHELD', frantic cuts, chase sequences, or similar.
- Drone/aerial: add if 'aerial', 'bird's eye view', 'drone', or overhead establishing shots are described. Flag: requires FAA-licensed operator.
- Underwater housing: add if any underwater or submerged scenes. Also triggers Stunts & Safety row.
- High-speed/slow-motion camera: add if script describes action in apparent slow motion or extreme detail that implies high frame rate.
- Second camera: add if simultaneous coverage of action is implied (car chases, crowd scenes, stunts).

2700 — Electric / Lighting

- Night scenes: add 'Night Lighting Package' row for every scene with NIGHT in slug. Night = automatic cost flag.
- Large exterior day scenes with crowds: add 'Exterior Lighting Package' — natural light often requires supplemental HMIs.
- Fog/smoke/haze: add 'Fog machine + fluid' row in 2700. Also check if it belongs in 3000 (Mechanical FX).
- Practical streetlights, neon signs, or specific atmospheric practicals scripted on camera: add as separate line items.
- Rain: add 'Rain tower / rain bar kit' row. Coordinate with 3000 (Mechanical FX) and 3500 (Locations — drainage).

3000 — Mechanical FX (Practical On-Set Effects)

- Fire / flames / burning: add 'Practical fire rig' + 'Fire safety officer'. High priority flag. May also need VFX row.
- Explosion / gunshot / crash impact: add 'Practical explosion rig' + 'Pyrotechnician'. Always add safety officer row.
- Rain: add 'Rain tower system'. Note drainage requirements for Locations department.
- Fog / smoke / mist: add 'Fogger + fluid'. If supernatural fog, may be VFX instead — flag for director decision.
- Snow: add 'Artificial snow materials'. If exterior, also flag Locations for weather contingency day.
- Bullet hits / squibs: add 'Squib system + operator' for each instance of on-screen gunshot wounds.

3100 — Special VFX (Visual Effects)

- Count estimated VFX shots from the script. Add one row per VFX category (digital environment, creature, invisible VFX, etc.).
- Green screen / LED volume: add 'VFX Plate Shoot' row and 'VFX Supervision' row.
- Wire work removal: add 'Digital wire removal' as VFX line item.
- Any scene described as impossible to achieve practically: flag as VFX and set Confidence=Medium.
- Miniatures: add in 3100 if miniature photography is implied. Cross-reference with 2600 (special camera).

3300 — Wardrobe

- One row per principal character's costume. Specify: period vs. contemporary, Purchase vs. Rental.
- Period costumes (any era other than present day): Unit = Rental-Day, flag for costume designer decision.
- Extras/background wardrobe: add as Allow row if crowd scenes exist. Period extras = much higher cost, flag separately.
- Quick-change scenes: add 'Quick-change wardrobe rigging' row with note about number of changes.
- Specialty wardrobe (military, medical, religious, uniforms): flag for research and sourcing time.
- Stunt doubles wardrobe: add matching wardrobe rows for any stunt double characters.

3400 — Makeup & Hair

- Standard makeup is NOT extracted from script — it is a formula cost based on cast size, handled by the Line Producer.
- Only extract NOTABLE makeup: prosthetics, aging, wounds, blood, creature makeup, extreme transformations.
- Prosthetics: add 'Prosthetic design + fabrication' row and 'Prosthetic application (per shoot day)' row separately.
- Aging makeup: add 'Aging makeup design' + 'Application (per day)' rows.
- Blood / wound gags: add 'Blood effects + supplies' row. Quantity = estimated number of makeup resets needed.
- Wigs: add if character has a wig as part of their established look.

3500 — Locations

- One row per distinct location in the script. Label = exact location name from slug line.
- Columns: C = number of shoot days at this location [SCHEDULE], D = Days, E = [DEAL].

- Add 'Location Permit' sub-row for every EXT. public space location (streets, parks, landmarks).
- Night exterior: add 'Night shoot surcharge' row for each night exterior location.
- Multiple days at same location: one row with [SCHEDULE] in the day count — do not estimate shoot days.
- Unique/high-demand locations (airports, hospitals, stadiums): flag with Note: 'High permit complexity — early scout required'.

3600 — Transportation (Picture Vehicles only)

- Claude only populates picture vehicle rows — vehicles that appear on screen.
- One row per vehicle type. Description = vehicle type (e.g. '1967 Ford Mustang', 'Police cruiser', 'Ambulance').
- Stunt vehicles: add separate row with note 'Stunt vehicle — duplicate of hero vehicle for action sequences'.
- Production vehicles (trucks, vans, honeywagons) are [SCHEDULE] placeholders only — do not extract from script.
- Helicopter / airplane on screen: add in 3600 with high priority flag. Also add FAA compliance note.

3900 — Atmosphere (Background Extras)

- One row per crowd scene. Description = scene location. Quantity = estimated extras count.
- If script specifies a number ('forty mourners', 'packed bar'): use that number, Confidence=High.
- If script implies a crowd but no number ('busy street', 'crowded marketplace'): estimate conservatively, Confidence=Low.
- Minors in crowd: flag separately. Add 'Welfare Worker/Teacher' row in 1600 if minors appear.
- Dancers or performers in crowd: add in 3900 as separate row. Different rate from general extras.

Stunts & Safety — rows distributed across 2500, 3000, 3600

- Stunt Coordinator: add one row in the account most relevant to the primary stunt type. If multiple stunt types, add in each relevant account.
- Physical contact between actors (fights, grabs, falls): always add Stunt Coordinator row, Confidence=High.
- Prop weapons (any edged weapon or firearm, real or prop): add Licensed Armorer row in 2500.
- Fire/explosion stunts: add Safety Officer row in 3000 in addition to stunt coordinator.
- Vehicle stunts: add Stunt Driver row in 3600 in addition to stunt coordinator.
- Children in physically demanding scenes: add Welfare Worker row in 1600 and Safety Officer row.

8. CLAUDE API — JSON OUTPUT FORMAT

Claude must return a single JSON object with two top-level keys: 'scene_breakdown' (array of scene objects) and 'accounts' (object keyed by account number, each containing an array of line item objects). No prose, no markdown fences, no explanation — raw JSON only.

JSON Schema:

```
{
  "scene_breakdown": [
    {
      "scene_number": "14",
      "heading": "INT. DINER - DAY",
      "int_ext": "INT",
      "day_night": "DAY",
      "page_count_eighths": "2/8",
      "script_day": "Day 3",
      "cast": ["Jake", "Maria"],
      "extras_count": 8,
      "extras_description": "Diner regulars",
      "props": ["Jakes gun", "Coffee cups"],
      "practical_fx": [],
      "wardrobe_notes": "",
      "animals": [],
      "picture_vehicles": [],
      "special_makeup": [],
      "special_camera": [],
      "stunts": ["Jake draws weapon"],
      "vfx": [],
      "location_type": "Practical",
      "flags": ["Prop weapon - armorer required"],
      "script_page": 12
    }
  ],
  "accounts": {
    "1600": [
      {
        "account_no": "1600.01",
        "description": "Principal Cast - Jake",
        "amount": 8,
        "unit_type": "Days",
        "populated_by": "SCRIPT",
        "script_page": 1,
        "script_quote": "JAKE enters, scanning the room.",
        "notes": "Appears in 8 of 12 scenes",
        "confidence": "High"
      }
    ],
    "2500": [
      {
        "account_no": "2500.01",
        "description": "Hero prop - Jakes gun (prop firearm)",
        "amount": 2,
        "unit_type": "Purchase",
        "populated_by": "SCRIPT",
        "script_page": 12,
        "script_quote": "Jake reaches for his gun under the table.",
        "notes": "Licensed armorer required. Two units: hero + backup.",
        "confidence": "High"
      }
    ]
  }
}
```

IMPORTANT: Amount for cast rows = number of scenes that character appears in (count from scene_breakdown array). This gives the Line Producer a scene count, not a day count — the day count comes from the shooting schedule. Claude must never guess shooting days.

9. CLAUDE API SYSTEM PROMPT — EMBED VERBATIM

The following system prompt must be embedded verbatim in the Python script's API call. Claude has no memory between runs — the rules must travel with every call.

You are an expert film production breakdown assistant with the knowledge of an experienced 1st Assistant Director and Line Producer. You read screenplay PDFs and extract structured production data for budgeting purposes.

OUTPUT RULES:

- Return ONLY a valid JSON object. No prose, no explanation, no markdown fences.
- Never hallucinate or guess. If a field cannot be determined, use null or empty array.
- Never estimate shooting days – leave day counts for the scheduling team.
- For cast Amount field: count the number of scenes each character appears in.

EXTRACTION RULES:

- Read every line of action and scene description – cost drivers are often implied not stated.
- One script element can generate multiple account rows (car chase = Transportation + Stunts + Camera).
- Period settings apply to ALL items in that scene: wardrobe, props, set dressing all need period versions.
- Dialogue alone rarely generates costs. Focus on action lines, scene headings, parentheticals.
- EXT. + public space = location permit row required.
- NIGHT in slug = night lighting package row required.
- Any weapon (prop or real) = armorer row required.
- Any animal = licensed handler row required (legally mandated).
- Any physical contact between actors = stunt coordinator row required.
- Any child actor = welfare worker/teacher row required.

CONFIDENCE LEVELS:

- High: item explicitly named in script.
- Medium: item strongly implied by scene description.
- Low: item inferred from atmosphere or subtext.

SCRIPT LANGUAGE LOOKUP:

fog/mist/haze/smoke -> Grip & Electric: fog machine + fluid
rain/downpour/wet streets -> Grip & Electric: rain tower kit
fire/flames/burning -> Stunts: fire rig + Mechanical FX: fire safety officer
crowd/hundreds/busy street -> Atmosphere: background extras (estimate qty)
period/specific decade -> Art Dept: period set dressing + Wardrobe: period costumes
police/ambulance/fire truck -> Transportation: picture vehicle rental
explosion/gunshot/crash -> Stunts: practical effect + safety coordinator
green screen/wire work -> VFX: supervision + plate shoot
animal/horse/dog -> Props: animal rental + licensed handler
drone/aerial/birds eye -> Camera: drone unit + FAA operator
underwater/submerged -> Camera: underwater housing + Stunts: dive safety
night (in slug) -> Electric: night lighting package
EXT + public space -> Locations: permit + fees
weapon/gun/knife -> Props: armorer + Stunts: safety officer
car chase/driving -> Transportation + Stunts + Camera (three accounts)
prosthetics/aging/wounds -> Makeup: special makeup FX
stunt double/falls/fight -> Stunts: coordinator + performer

10. BUILD ORDER FOR CLAUDE CODE

Step	Action	Verify
1	Update the Claude API call to use the new JSON schema (Section 8). Update the system prompt to the verbatim text in Section 9.	Test with a short script. Confirm JSON contains both 'scene_breakdown' and 'accounts' keys.
2	Build the Scene Breakdown tab writer. One row per scene object from JSON. Apply industry color coding to column headers (Section 5).	Open output. Confirm one row per scene. Confirm [SCHEDULE] placeholder in columns S and T.
3	Build the account tab template writer. For each of the 30+ account tabs, pre-fill the standard Movie Magic line item labels in column B as gray placeholder rows. These template rows exist even when Claude finds nothing in the script.	Open a blank-input run. Confirm all account tabs exist with standard labels pre-filled.
4	Build the account tab script-row writer. For each account in the JSON 'accounts' object, insert extracted rows above the template rows. Apply green fill to column O ('SCRIPT') for extracted rows.	Run with a real script. Confirm extracted rows appear in correct account tabs with green O column.
5	Write all Subtotal and Total columns as Excel formula strings. Subtotal = $=C\{r\}*D\{r\}*E\{r\}$. Total = $=F\{r\}+(F\{r\}*G\{r\})$. Never hardcode.	Open in Excel. Click a Total cell. Confirm it shows a formula string, not a hardcoded number.
6	Build the Top Sheet tab. One row per account group. Each total cell = SUM formula referencing column H of the corresponding account tab.	Manually enter a Rate in any account tab. Confirm Top Sheet grand total updates automatically.
7	Implement Script Page hyperlinks (column K on account tabs, column V on Scene Breakdown). Use openpyxl write_url() pointing to the --input PDF path.	Click a hyperlink in the output file. Confirm it opens the correct page of the source PDF.
8	Apply color coding: green fill for SCRIPT rows, yellow for SCHEDULE, pink for DEAL, blue for FORMULA in column O. Gray fill for all placeholder template rows.	Visual check: color bands clearly distinguish populated vs. manual-entry rows throughout all tabs.
9	End-to-end test with two contrasting scripts: (a) a minimal two-character one-location short film, (b) a complex action scene with stunts, crowds, vehicles, and night exteriors.	Simple script = lean output with few SCRIPT rows. Complex script = many extracted rows across multiple account tabs.
10	Freeze column headers on all tabs. Set appropriate column widths. Lock formula cells from manual editing. Save final file as .xlsx.	Final QA: producer opens file, manually enters a Rate value in any tab, Grand Total on Top Sheet updates correctly.

11. HANDOFF TO TOOL 2 (PRICING ENGINE)

Before building Tool 2, confirm the column letters of the integration contract with Claude Code. Tool 2 writes into:

- Column E (Rate) — Tool 2 populates this from live pricing APIs or vendor databases.
- Column I (Vendor) — Tool 2 writes the vendor name.
- Column J (Vendor URL) — Tool 2 writes a clickable hyperlink to the pricing source.

Columns E, I, and J must remain blank in the Tool 1 output. Tool 2 reads the existing breakdown file, iterates every account tab, finds rows where Populated By = SCRIPT and Rate is blank, and attempts to fetch a real-world price for that item. The formula in column F then calculates automatically.

Do not change column letters E, F, G, H, I, J after Tool 1 is built. These are the permanent integration contract. Any change requires updating both tools simultaneously.

Opening Prompt for Claude Code

Use the following prompt when starting the Claude Code session for this build:

I've attached a build spec PDF (Spec Sheet Version 2) for updating my script breakdown tool. Please read it carefully. This is an UPDATE to an existing tool – the CLI interface, API call architecture, and openpyxl generation are already built. The change is entirely to the output spreadsheet structure. The new output must use the Movie Magic account numbering system with one tab per account, a Scene Breakdown tab, and a Top Sheet summary tab.

Start by confirming your understanding of:

1. The three population types (SCRIPT / SCHEDULE / DEAL) and how they appear in the sheet
2. The universal column structure (A through O) used on every account tab
3. Which accounts Claude populates vs. which are template placeholders

Then proceed with the build order in Section 10 of the spec, one step at a time. Show me the output of each step before proceeding to the next.